SwarRakshak REST API Specification

1. API purpose

This document defines the REST API for the SwarRakshak hackathon MVP. The API accepts a short call sample and structured call context, runs the analysis pipeline, and returns an explainable impersonation-risk result.

The API is intentionally small. It supports the product's required workflow without adding unnecessary telecom, banking, notification, or administration endpoints.

Core workflow

```
Create analysis → Check analysis result → View flagged-analysis summaries
```

Important safety rule

The API provides a risk assessment and recommendation. It does **not** approve or reject a payment, block a call, or execute a financial transaction.

2. API conventions

Base URL

For local development:

```
http://localhost:8000/api/v1
```

For a deployed environment, replace the host with the application's HTTPS domain.

Content types

- `multipart/form-data` for creating an analysis because the request may include audio files.
- `application/json` for regular responses and query parameters.

- `application/json` for errors.

Identifier format

The API uses text identifiers generated by the backend:

```
usr_<unique-value>
evt_<unique-value>
```

Time format

All timestamps use ISO 8601 UTC format:

```
2026-09-16T14:30:10Z
```

Risk values

- `low` : 0–29
- `medium` : 30–59
- `high` : 60–100
- `inconclusive` : analysis evidence is insufficient

These thresholds are prototype settings and must be configurable in the application.

3. Authentication and authorization

Hackathon mode

For a local demo, authentication may be disabled or replaced with a fixed demo user. The API should still accept an optional `X-Demo-User-Id` header if the team wants to associate events with a user.

Recommended deployed mode

Use a bearer token:

```
Authorization: Bearer <access-token>
```

The token should identify the user and role. Do not build a custom password system for the hackathon unless the team already has one.

Role requirements

Role	Allowed operations
`finance_user`	Create an analysis and view its own analysis result
`fraud_analyst`	Create analyses and view analysis summaries and details
`admin`	All prototype API operations

Authentication error

If authentication is enabled and the token is absent or invalid, return:

```
{
  "error": {
    "code": "AUTHENTICATION_REQUIRED",
    "message": "A valid access token is required."
  }
}
```

4. Common error format

All error responses should use the same structure.

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "One or more request fields are invalid.",
    "details": [
      {
        "field": "request_urgency",
        "message": "Value must be 'normal' or 'urgent'."
      }
    ],
    "request_id": "req_01J9SWR999"
  }
}
```

Error fields

Field	Required	Description
`error.code`	Yes	Stable machine-readable code
`error.message`	Yes	Safe human-readable message
`error.details`	No	Field-level validation information
`error.request_id`	Recommended	ID for debugging and support logs

The API must not return stack traces, database paths, temporary file names, model internals, or sensitive audio information in an error response.

5. Endpoint summary

Method	URL	Purpose	Authentication
`GET`	`/health`	Check API and model availability	None for MVP
`POST`	`/analyses`	Create and run a call analysis	Optional in demo; required in deployment
`GET`	`/analyses/{event_id}`	Get one analysis result	Owner, fraud analyst, or admin
`GET`	`/analyses`	List analysis summaries for investigation	Fraud analyst or admin

No endpoint is provided for approving payments, blocking calls, sending external alerts, or storing raw audio because those actions are outside the hackathon MVP.

6. Endpoint specifications

6.1 Health check

HTTP method and URL

```
GET /api/v1/health
```

Purpose

Checks whether the API is running and whether the analysis components are available.

Authentication requirement

No authentication is required for the hackathon health check. In a production deployment, the endpoint may be protected or split into a basic liveness endpoint and an authenticated readiness endpoint.

Request parameters

None.

Request body

None.

Example request

```
curl -X GET http://localhost:8000/api/v1/health
```

Example success response

```
{
  "status": "ok",
  "service": "swarrakshak-api",
  "version": "0.1.0",
  "audio_model": "ready",
  "speaker_model": "ready",
  "checked_at": "2026-09-16T14:30:00Z"
}
```

Response fields

Field	Type	Description
`status`	String	`ok` or `degraded`
`service`	String	Service name
`version`	String	API version or build version
`audio_model`	String	`ready`, `unavailable`, or `not_configured`
`speaker_model`	String	`ready`, `unavailable`, or `not_configured`
`checked_at`	String	UTC timestamp

Possible error responses

503 Service Unavailable

```
{
  "error": {
    "code": "SERVICE_UNAVAILABLE",
    "message": "The API is running, but one or more required analysis com
  }
}
```

HTTP status codes

Status	Meaning
`200`	API is running and usable
`503`	API or required analysis dependency is unavailable

6.2 Create an analysis

HTTP method and URL

```
POST /api/v1/analyses
```

Purpose

Creates an analysis event, accepts a current-call audio sample, optionally accepts a trusted reference sample, analyzes the available evidence, applies call-context rules, and returns a risk result.

The hackathon may process the request synchronously. If processing later becomes slow, the same endpoint can return `202 Accepted` and expose a processing status through the get-analysis endpoint.

Authentication requirement

- **Hackathon local mode:** Optional; associate the request with a demo user.
- **Deployed mode:** Required. The caller must have the `finance_user`, `fraud_analyst`, or `admin` role.

Request content type

```
Content-Type: multipart/form-data
```

Request parameters and form fields

Field	Type	Required	Description
`current_audio`	File	Yes	Current call sample in WAV, MP3, or M4A format
`reference_audio`	File	No	Consented sample of the person the caller claims to be
`caller_number_status`	String	Yes	`known`, `unknown`, or `hidden`
`request_urgency`	String	Yes	`normal` or `urgent`
`transaction_value`	String	Yes	`low`, `medium`, or `high`
`procedure_status`	String	Yes	`normal` or `bypass_requested`
`caller_relationship`	String	Yes	`customer`, `vendor`, `executive`, or `unknown`
`retain_audio`	Boolean	No	Defaults to `false`; retaining audio should be disabled or restricted in the MVP

File constraints

The implementation should define and document exact limits. Suggested hackathon limits are:

- Maximum file size: 25 MB per file.
- Maximum current-audio duration: 60 seconds.

- Maximum reference-audio duration: 60 seconds.
- Supported formats: WAV, MP3, and M4A.
- Audio must contain a decodable speech signal.

The backend must validate the decoded audio rather than trusting only the file extension.

Request body

Because this is a multipart request, fields are sent as form parts rather than as a single JSON object.

Example request

```
curl -X POST http://localhost:8000/api/v1/analyses \
  -H "Authorization: Bearer <access-token>" \
  -F "current_audio=@samples/current_call.wav" \
  -F "reference_audio=@samples/cfo_reference.wav" \
  -F "caller_number_status=unknown" \
  -F "request_urgency=urgent" \
  -F "transaction_value=high" \
  -F "procedure_status=bypass_requested" \
  -F "caller_relationship=executive" \
  -F "retain_audio=false"
```

Processing behavior

The backend should perform these steps:

1. Create an event ID.
2. Validate the request and uploaded files.
3. Store audio temporarily outside the public web directory.
4. Convert and segment the audio.
5. Run voice-authenticity analysis.
6. Run speaker comparison if a reference sample exists.
7. Apply context and policy rules.
8. Calculate a 0–100 risk score.
9. Generate reasons and a recommended action.

10. Store metadata and delete raw audio by default.
11. Return the result.

Example success response: completed analysis


```
{
  "event_id": "evt_01J9SWR123",
  "status": "completed",
  "created_at": "2026-09-16T14:30:00Z",
  "completed_at": "2026-09-16T14:30:07Z",
  "risk_score": 82,
  "risk_level": "high",
  "confidence_level": "high",
  "confidence_score": 0.78,
  "evidence_sufficient": true,
  "audio_analysis": {
    "available": true,
    "synthetic_probability": 0.86
  },
  "speaker_analysis": {
    "available": true,
    "similarity": 0.28,
    "confidence_score": 0.69
  },
  "context": {
    "caller_number_status": "unknown",
    "request_urgency": "urgent",
    "transaction_value": "high",
    "procedure_status": "bypass_requested",
    "caller_relationship": "executive"
  },
  "reasons": [
    {
      "code": "SYNTHETIC_AUDIO_PATTERN",
      "message": "Audio contains patterns associated with synthetic speecl
      "source": "audio",
      "contribution": 40
    },
    {
      "code": "REFERENCE_VOICE_MISMATCH",
      "message": "Current voice differs from the reference sample.",
      "source": "speaker",
      "contribution": 30
    },
    {
      "code": "UNKNOWN_CALLER_NUMBER",
```

```
      "message": "Caller number is not recognized as trusted.",
      "source": "context",
      "contribution": 10
    },
    {
      "code": "URGENT_OR_BYPASS_REQUEST",
      "message": "Caller is creating urgency or asking to bypass procedur
      "source": "context",
      "contribution": 10
    }
  ],
  "segments": [
    {
      "segment_number": 1,
      "start_seconds": 0.0,
      "end_seconds": 4.0,
      "synthetic_probability": 0.81,
      "confidence_score": 0.72,
      "status": "analyzed"
    },
    {
      "segment_number": 2,
      "start_seconds": 4.0,
      "end_seconds": 8.0,
      "synthetic_probability": 0.91,
      "confidence_score": 0.76,
      "status": "analyzed"
    }
  ],
  "recommended_action": "Pause the transfer and call the person using an
  "raw_audio_retained": false,
  "audio_deleted_at": "2026-09-16T14:30:08Z",
  "model_version": "prototype-audio-1",
  "rules_version": "risk-rules-1"
}
```

Example success response: inconclusive analysis


```
{
  "event_id": "evt_01J9SWR124",
  "status": "completed",
  "risk_score": 20,
  "risk_level": "inconclusive",
  "confidence_level": "low",
  "confidence_score": 0.18,
  "evidence_sufficient": false,
  "audio_analysis": {
    "available": false,
    "synthetic_probability": null
  },
  "speaker_analysis": {
    "available": false,
    "similarity": null,
    "confidence_score": null
  },
  "reasons": [
    {
      "code": "INSUFFICIENT_AUDIO_EVIDENCE",
      "message": "The audio is too short or noisy for a reliable analysis
      "source": "system",
      "contribution": 0
    }
  ],
  "recommended_action": "Treat the request as unverified and use an indep
  "raw_audio_retained": false,
  "model_version": "prototype-audio-1",
  "rules_version": "risk-rules-1"
}
```

Possible error responses

400 Bad Request — invalid form value

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "One or more request fields are invalid.",
    "details": [
      {
        "field": "transaction_value",
        "message": "Value must be 'low', 'medium', or 'high'."
      }
    ],
    "request_id": "req_01J9SWR001"
  }
}
```

400 Bad Request — missing audio

```
{
  "error": {
    "code": "CURRENT_AUDIO_REQUIRED",
    "message": "The current_audio file is required."
  }
}
```

413 Payload Too Large

```
{
  "error": {
    "code": "AUDIO_FILE_TOO_LARGE",
    "message": "The uploaded audio exceeds the maximum allowed size."
  }
}
```

415 Unsupported Media Type


```
{
  "error": {
    "code": "UNSUPPORTED_AUDIO_FORMAT",
    "message": "Supported audio formats are WAV, MP3, and M4A."
  }
}
```


422 Unprocessable Entity


```
{
  "error": {
    "code": "AUDIO_NOT_DECODABLE",
    "message": "The audio file could not be decoded or does not contain a
  }
}
```


500 Internal Server Error


```
{
  "error": {
    "code": "ANALYSIS_FAILED",
    "message": "The analysis could not be completed. Please try again.",
    "request_id": "req_01J9SWR002"
  }
}
```

503 Service Unavailable

```
{
  "error": {
    "code": "ANALYSIS_MODEL_UNAVAILABLE",
    "message": "The required analysis model is temporarily unavailable."
  }
}
```

HTTP status codes

Status	Meaning
`200`	Analysis completed and result returned
`202`	Analysis accepted for asynchronous processing, if that mode is implemented
`400`	Invalid form values or missing required input
`401`	Authentication required or token invalid
`403`	Authenticated user lacks permission
`413`	Uploaded file is too large
`415`	File format is unsupported
`422`	File cannot be decoded or analyzed
`500`	Unexpected analysis failure
`503`	Required analysis dependency is unavailable

6.3 Get one analysis result

HTTP method and URL

```
GET /api/v1/analyses/{event_id}
```

Purpose

Returns the metadata and analysis result for one event. This endpoint supports the result page and asynchronous processing if the team adds it later.

It must not return raw audio by default.

Authentication requirement

Required in deployed mode.

- A `finance_user` may view an event they created.
- A `fraud_analyst` or `admin` may view events allowed by the organization.
- In local demo mode, the endpoint may be unauthenticated.

Path parameters

Parameter	Type	Required	Description
`event_id`	String	Yes	Event identifier, such as `evt_01J9SWR123`

Request body

None.

Example request

```
curl -X GET \
  http://localhost:8000/api/v1/analyses/evt_01J9SWR123 \
  -H "Authorization: Bearer <access-token>"
```

Example success response

The response has the same result structure as the completed `POST /analyses` response. A shorter example is shown below.

```
{
  "event_id": "evt_01J9SWR123",
  "status": "completed",
  "created_at": "2026-09-16T14:30:00Z",
  "completed_at": "2026-09-16T14:30:07Z",
  "risk_score": 82,
  "risk_level": "high",
  "confidence_level": "high",
  "confidence_score": 0.78,
  "evidence_sufficient": true,
  "reasons": [
    {
      "code": "SYNTHETIC_AUDIO_PATTERN",
      "message": "Audio contains patterns associated with synthetic speecl
      "source": "audio",
      "contribution": 40
    }
  ],
  "recommended_action": "Pause the transfer and call the person using an 
  "raw_audio_retained": false
}
```


Example response while processing


```
{
  "event_id": "evt_01J9SWR125",
  "status": "processing",
  "created_at": "2026-09-16T14:31:00Z",
  "risk_score": null,
  "risk_level": null,
  "message": "The audio is still being analyzed."
}
```

Possible error responses

400 Bad Request — invalid event ID

```
{
  "error": {
    "code": "INVALID_EVENT_ID",
    "message": "The event_id format is invalid."
  }
}
```

401 Unauthorized

Use the common authentication error format.

403 Forbidden

```
{
  "error": {
    "code": "FORBIDDEN",
    "message": "You do not have permission to view this analysis."
  }
}
```

404 Not Found

```
{
  "error": {
    "code": "ANALYSIS_NOT_FOUND",
    "message": "No analysis was found for the supplied event_id."
  }
}
```

HTTP status codes

Status	Meaning
`200`	Analysis found and returned
`400`	Invalid event ID
`401`	Authentication required or invalid
`403`	User is not allowed to view the event
`404`	Event does not exist
`500`	Unexpected database or serialization error

6.4 List analysis summaries

HTTP method and URL

```
GET /api/v1/analyses
```

Purpose

Returns a paginated list of metadata-only analysis summaries. This supports the fraud analyst user story: reviewing flagged events without immediately listening to recordings.

The endpoint must not return raw audio or full call transcripts.

Authentication requirement

Required in deployed mode. Only `fraud_analyst` and `admin` roles should use this endpoint. A `finance_user` should use the single-analysis endpoint for their own events instead.

Query parameters

Parameter	Type	Required	Default	Description
`risk_level`	String	No	All	`low`, `medium`, `high`, or `inconclusive`
`status`	String	No	`completed`	`started`, `processing`, `completed`, or `failed`
`created_from`	String	No	None	ISO 8601 UTC start time
`created_to`	String	No	None	ISO 8601 UTC end time
`page`	Integer	No	`1`	Page number, minimum 1
`page_size`	Integer	No	`20`	Number of records, maximum 100
`sort`	String	No	`created_at_desc`	`created_at_desc` or `risk_score_desc`

Request body

None.

Example request

```
curl -G http://localhost:8000/api/v1/analyses \
  -H "Authorization: Bearer <access-token>" \
  --data-urlencode "risk_level=high" \
  --data-urlencode "page=1" \
  --data-urlencode "page_size=20"
```

Example success response


```
{
  "items": [
    {
      "event_id": "evt_01J9SWR123",
      "status": "completed",
      "created_at": "2026-09-16T14:30:00Z",
      "risk_score": 82,
      "risk_level": "high",
      "confidence_level": "high",
      "top_reasons": [
        "Audio contains patterns associated with synthetic speech.",
        "Caller number is not recognized as trusted."
      ],
      "recommended_action": "Pause the transfer and independently verify
      "raw_audio_retained": false
    }
  ],
  "pagination": {
    "page": 1,
    "page_size": 20,
    "total_items": 1,
    "total_pages": 1
  }
}
```


Possible error responses

400 Bad Request — invalid filter


```
{
  "error": {
    "code": "INVALID_QUERY_PARAMETER",
    "message": "page_size must be between 1 and 100."
  }
}
```

401 Unauthorized

Use the common authentication error format.

403 Forbidden


```
{
  "error": {
    "code": "ANALYST_ACCESS_REQUIRED",
    "message": "Only fraud analysts and administrators can list analysis s
  }
}
```


HTTP status codes

Status	Meaning
`200`	Summaries returned
`400`	Invalid query parameter
`401`	Authentication required or invalid
`403`	User lacks analyst access
`500`	Unexpected database error

7. Shared response objects

7.1 Reason object

```
{
  "code": "SYNTHETIC_AUDIO_PATTERN",
  "message": "Audio contains patterns associated with synthetic speech.",
  "source": "audio",
  "contribution": 40
}
```

Field	Type	Description
`code`	String	Stable reason code
`message`	String	User-facing explanation
`source`	String	`audio`, `speaker`, `context`, or `system`
`contribution`	Integer	Points added to the risk score

7.2 Segment object

```
{
  "segment_number": 1,
  "start_seconds": 0.0,
  "end_seconds": 4.0,
  "synthetic_probability": 0.81,
  "confidence_score": 0.72,
  "status": "analyzed"
}
```

The values `synthetic_probability` and `confidence_score` may be `null` when the segment is inconclusive.

7.3 Context object

```
{
  "caller_number_status": "unknown",
  "request_urgency": "urgent",
  "transaction_value": "high",
  "procedure_status": "bypass_requested",
  "caller_relationship": "executive"
}
```

7.4 Analysis result object

The completed-analysis response should contain:

Field	Type	Description
`event_id`	String	Unique analysis event
`status`	String	Lifecycle status
`created_at`	String	Start time
`completed_at`	String	Completion time, if completed
`risk_score`	Integer or null	Score from 0 to 100
`risk_level`	String or null	Low, Medium, High, or Inconclusive
`confidence_level`	String	Low, Medium, or High
`confidence_score`	Number or null	Value from 0.0 to 1.0
`evidence_sufficient`	Boolean	Whether evidence was sufficient
`audio_analysis`	Object	Audio-analysis summary
`speaker_analysis`	Object	Speaker-comparison summary
`context`	Object	Submitted context
`reasons`	Array	Explanation objects
`segments`	Array	Segment results, if available
`recommended_action`	String	Suggested next step
`raw_audio_retained`	Boolean	Whether audio was retained
`audio_deleted_at`	String or null	Deletion time, if deleted
`model_version`	String	Model version
`rules_version`	String	Risk-rule version

8. Validation rules

The backend must enforce these rules before analysis:

- `current_audio` is required.
- `reference_audio` is optional.
- Audio file type, decoded format, size, and duration must be validated.
- `caller_number_status` must be `known`, `unknown`, or `hidden`.
- `request_urgency` must be `normal` or `urgent`.
- `transaction_value` must be `low`, `medium`, or `high`.
- `procedure_status` must be `normal` or `bypass_requested`.
- `caller_relationship` must be `customer`, `vendor`, `executive`, or `unknown`.
- `retain_audio` defaults to `false`.
- `risk_score` must be between 0 and 100.
- Probability and confidence values must be between 0.0 and 1.0.
- `page` must be at least 1.
- `page_size` must be between 1 and 100.

9. Privacy and security requirements

9.1 Raw audio handling

- Do not return raw audio from any endpoint.
- Do not store raw audio in the database.
- Store uploaded files temporarily outside the public web directory.
- Delete raw audio after analysis by default.
- Return `raw_audio_retained: false` when deletion is complete.

9.2 Sensitive information

The API must not accept or store passwords, financial credentials, one-time passcodes, or complete confidential transcripts as part of the analysis context.

9.3 Access control

A user must not be able to retrieve another user’s analysis unless their role permits investigation access. The backend must enforce this rule rather than relying only on the frontend.

9.4 Rate limiting

A production deployment should rate-limit file uploads and analysis requests. For the hackathon, a simple limit such as one active analysis per demo user is sufficient.

9.5 Logging

Logs may contain event IDs, status, processing time, and safe error codes. Logs must not contain raw audio, access tokens, or sensitive request contents.

10. Recommended implementation order

1. Implement `GET /health`.
2. Implement `POST /analyses` with a fixed demo response.
3. Connect file validation and temporary audio cleanup.
4. Add the context-based scoring rules.
5. Add the real or prototype audio-analysis component.
6. Add `GET /analyses/{event_id}`.
7. Add `GET /analyses` for the fraud-analyst dashboard.
8. Add authentication only if the core flow is already working.
9. Test the three prepared demo scenarios.

11. Final API recommendation

The hackathon team should implement four endpoints only:

```
GET  /api/v1/health
POST /api/v1/analyses
GET  /api/v1/analyses/{event_id}
GET  /api/v1/analyses
```

This API is sufficient to demonstrate the complete SwarRakshak workflow while staying small, understandable, and safe. It supports analysis, result display, and fraud-analyst review without pretending to replace trusted identity verification or banking controls.